PMSCS 670

Software Testing

Graph Coverage

Lecture 7

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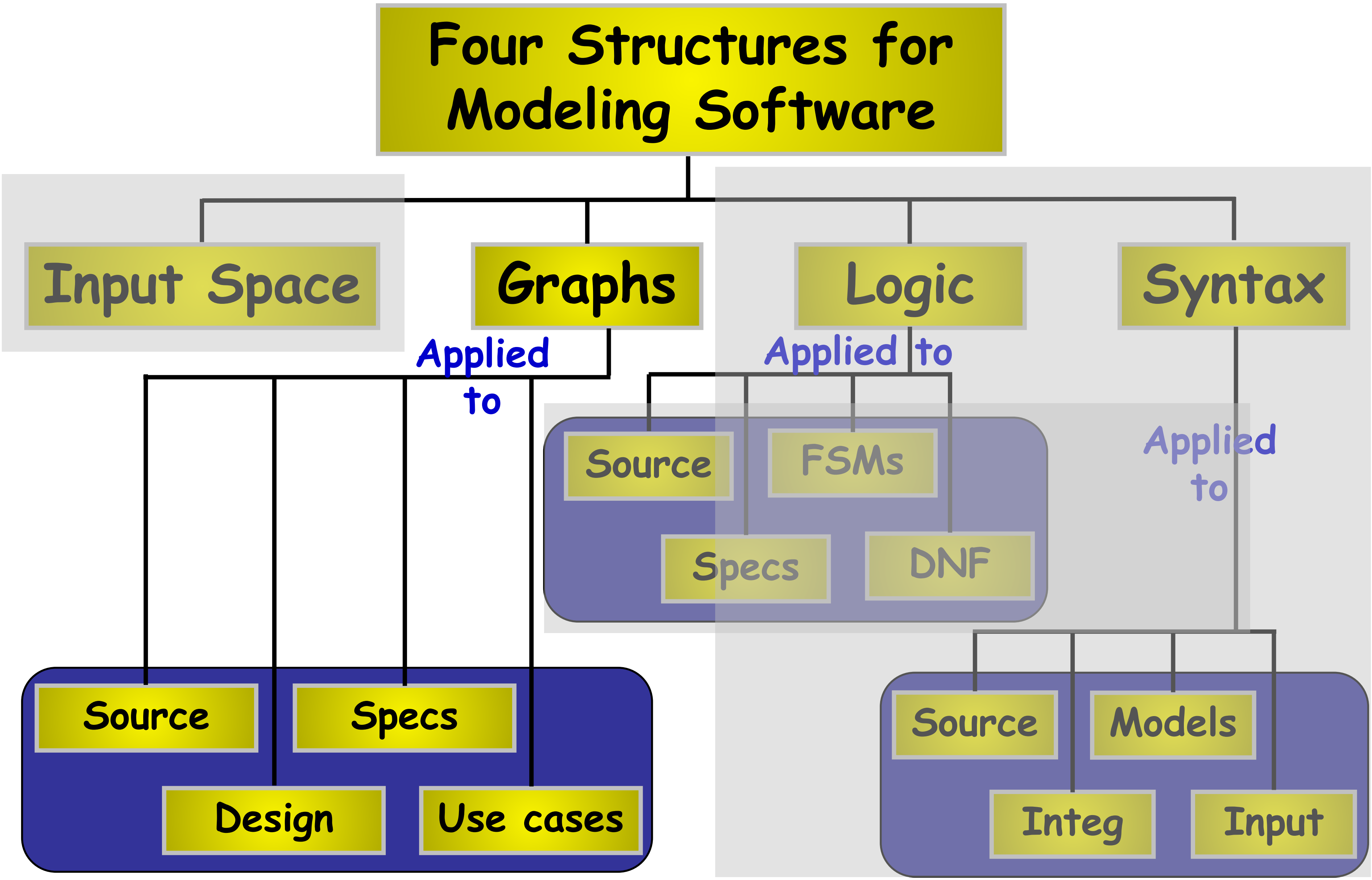
Ch. 7 : Graph Coverage

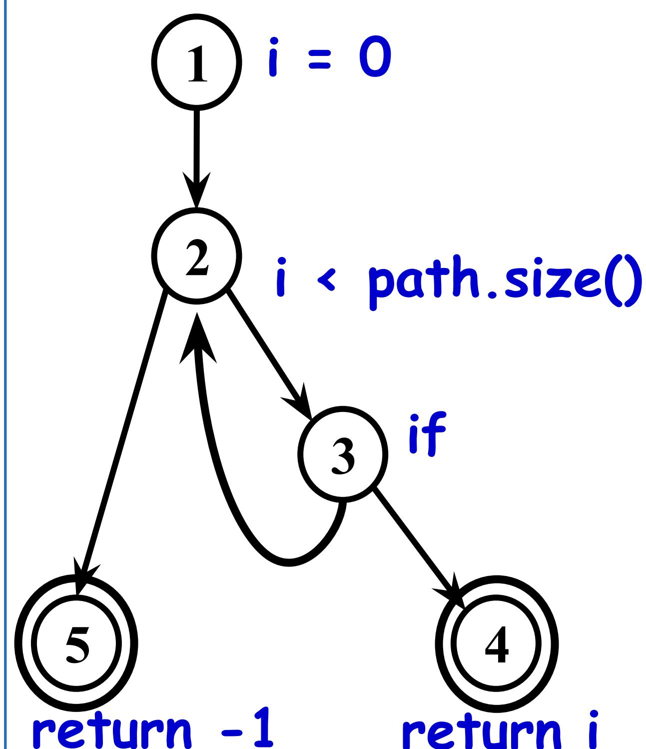


Small Illustrative Example

Software Artifact : Java Method

```
/**
   * Return index of node n at the
   * first position it appears,
   * -1 if it is not present
*/
public int indexOf (Node n)
{
   for (int i=0; i < path.size(); i++)
      if (path.get(i).equals(n))
         return i;
   return -1;
}
```

Control Flow Graph

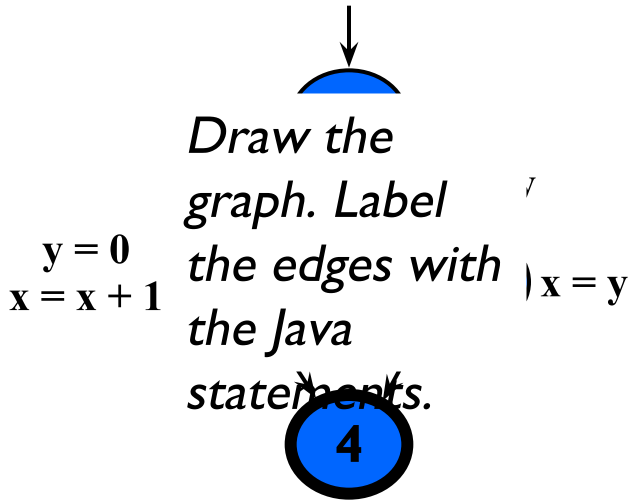


Control Flow Graphs

- A CFG models all executions of a method by describing control structures
- Nodes : Statements or sequences of statements (basic blocks)
- Edges : Transfers of control
- Basic Block : A sequence of statements such that if the first statement is executed, all statements will be (no branches)
- CFGs are sometimes annotated with extra information
  - branch predicates
  - defs
  - uses
- Rules for translating statements into graphs …

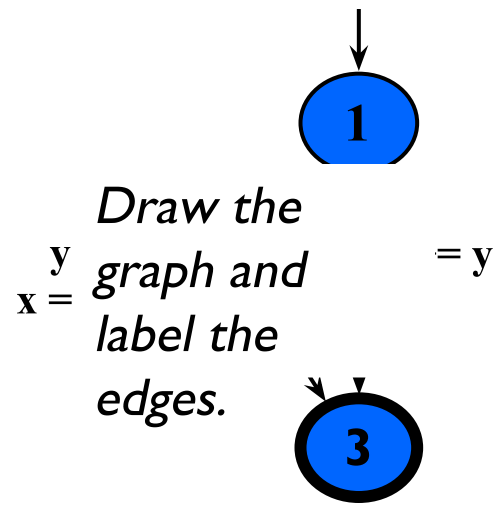
CFG : The if Statement

```
if (x < y)
{
   y = 0;
   x = x + 1;
}
else
{
   x = y;
}
```



Draw the graph. Label the edges with the Java statements.

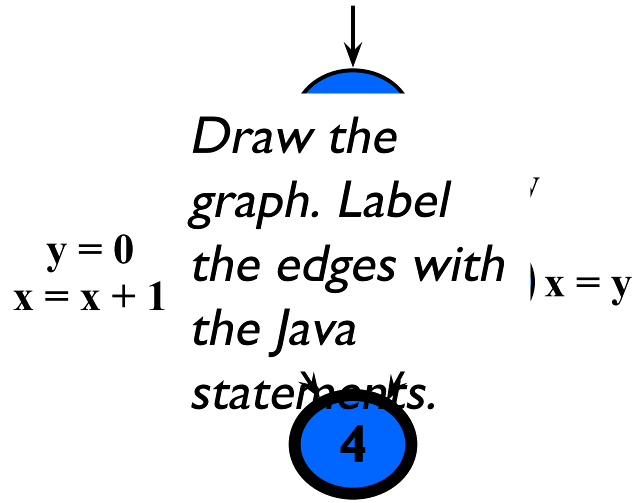
```
if (x < y)
{
   y = 0;
   x = x + 1;
}
```



Draw the graph and label the edges.

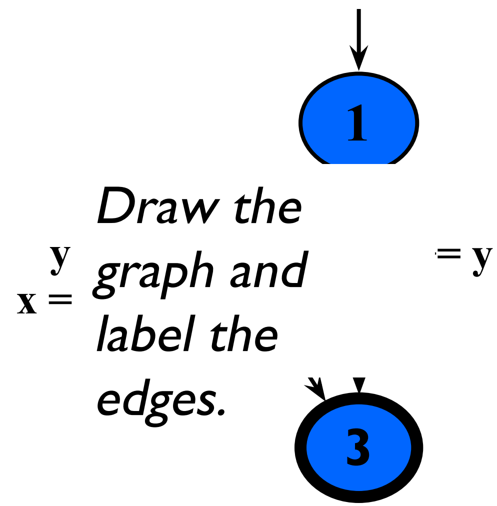
CFG : The if Statement

```
if (x < y)
{
  y = 0;
  x = x + 1;
}
else
{
  x = y;
}
```



Draw the graph. Label the edges with the Java statements.

```
if (x < y)
{
  y = 0;
  x = x + 1;
}
```

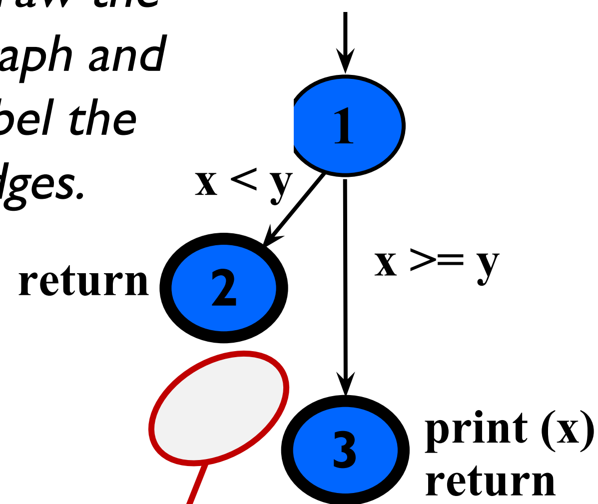


Draw the graph and label the edges.

CFG : The if-Return Statement

```
if (x < y)
{
   return;
}
print (x);
return;
```

Draw the graph and label the edges.



No edge from node 2 to 3.
The return nodes must be distinct.

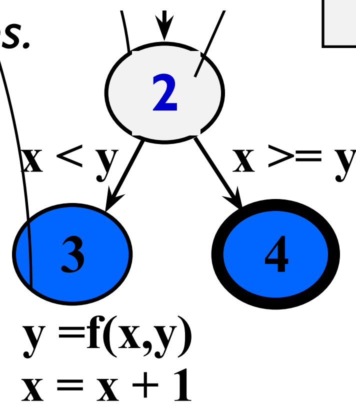
Loops

- Loops require "*extra*" nodes to be added
- Nodes that do not represent statements or basic blocks

CFG : while and for Loops

```
x = 0;
while (x < y)
{
   y = f (x, y);
   x = x + 1;
}
```

Draw the graph and label the edges.

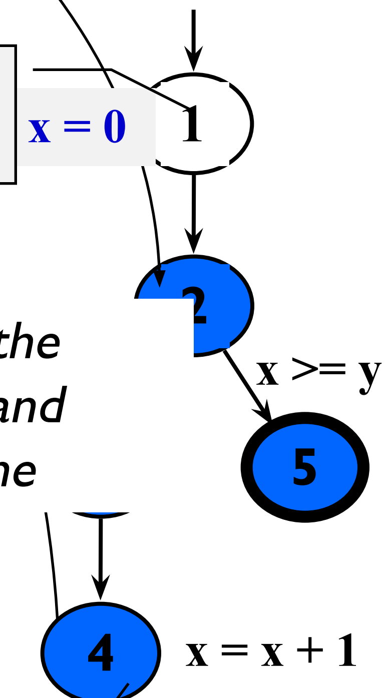


***dummy* node**

```
for (x = 0; x < y; x++)
{
   y = f (x, y);
}
```

implicitly initializes loop

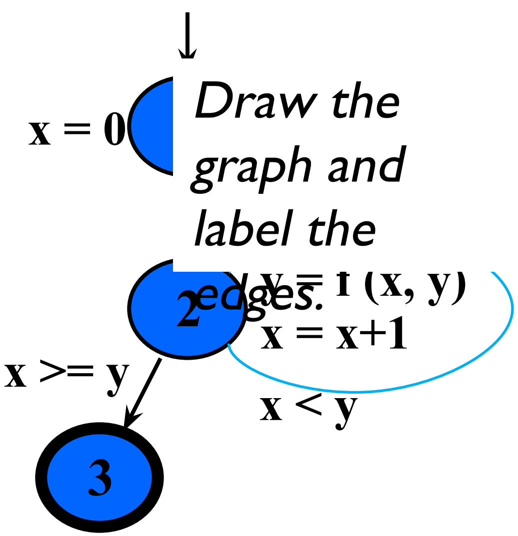
Draw the graph and label the edges.



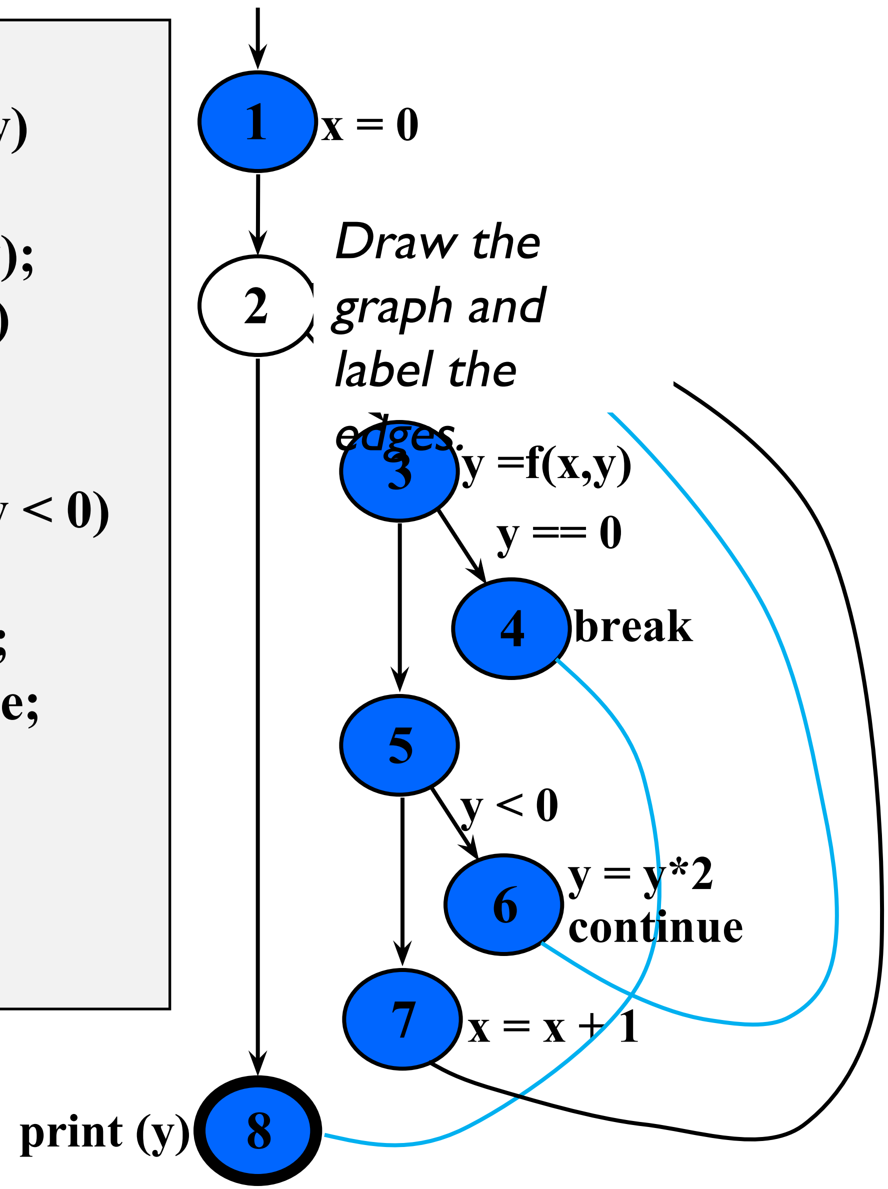
implicitly increments loop

CFG : do Loop, break and continue

```
x = 0;
do
{
   y = f (x, y);
   x = x + 1;
} while (x < y);
println (y)
```



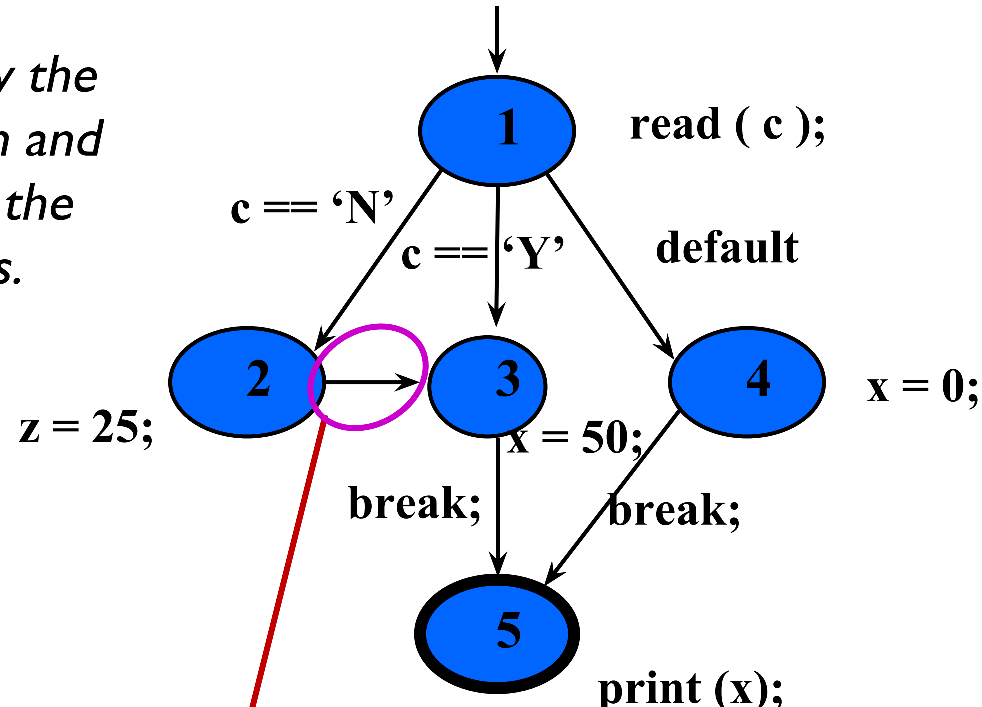

```
x = 0;
while (x < y)
{
   y = f (x, y);
   if (y == 0)
   {
      break;
   } else if (y < 0)
   {
      y = y*2;
      continue;
   }
   x = x + 1;
}
print (y);
```



CFG : The case (switch) Structure

```
read ( c) ;
switch ( c )
{
  case 'N':
    z = 25;
  case 'Y':
    x = 50;
    break;
  default:
    x = 0;
    break;
}
print (x);
```

Draw the graph and label the edges.

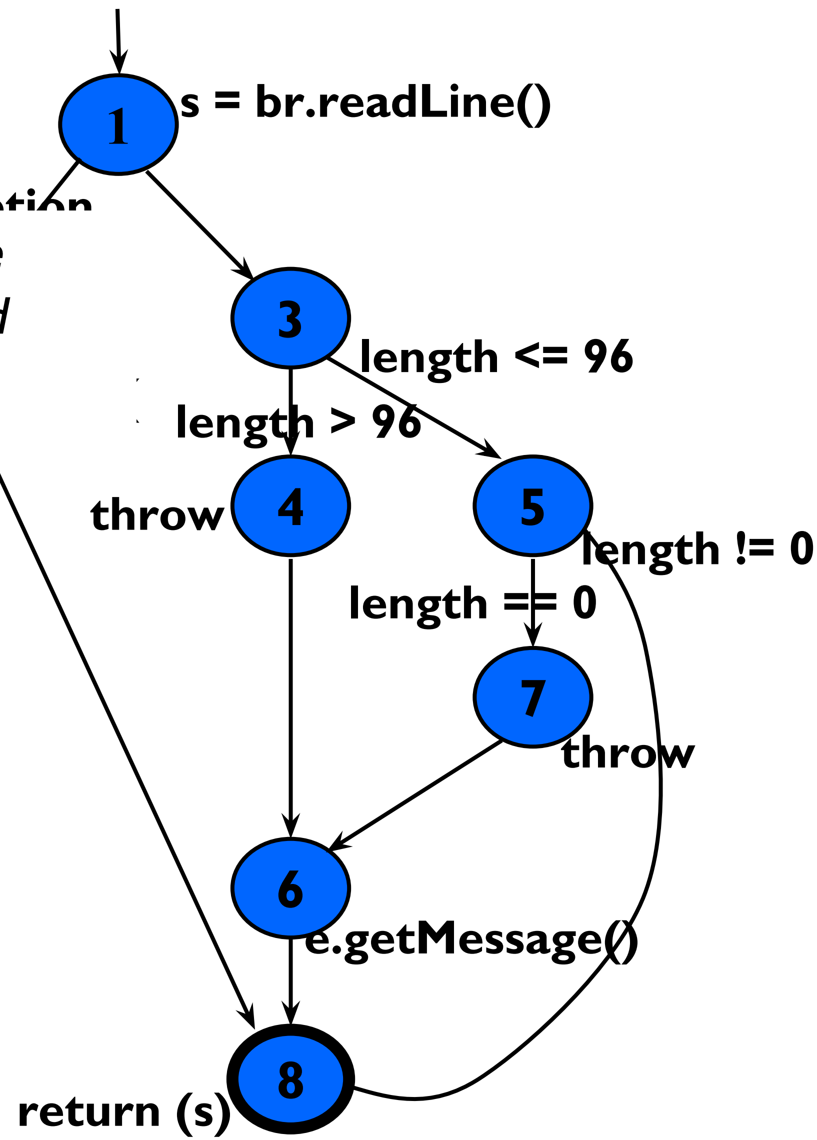


Cases without breaks fall through to the next case

CFG : Exceptions (try-catch)

```
try
{
   s = br.readLine();
   if (s.length() > 96)
      throw new Exception
         ("too long");
   if (s.length() == 0)
      throw new Exception
         ("too short");
} (catch IOException e) {
   e.printStackTrace();
} (catch Exception e) {
   e.getMessage();
}
return (s);
```

Draw the graph and label the edges.



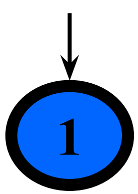
Covering Graphs (7.1)

- Graphs are the most commonly used structure for testing
- Graphs can come from many sources
  - Control flow graphs
  - Design structure
  - FSMs and statecharts
  - Use cases
- Tests usually are intended to "cover" the graph in some way

Definition of a Graph

- Any Graph $G(N, N_{0,} N_{f,} E)$, can be defined as-
  - A set N of nodes, N is not empty
  - A set N_0 of initial nodes, N_0 is not empty and $N_0 \subseteq N$
  - A set N_f of final nodes, N_f is not empty and $N_f \subseteq N$
  - A set E of edges, each edge from one node to another (n_i, n_j), where i is predecessor, j is successor $E \subseteq N x N$

Is this a graph?

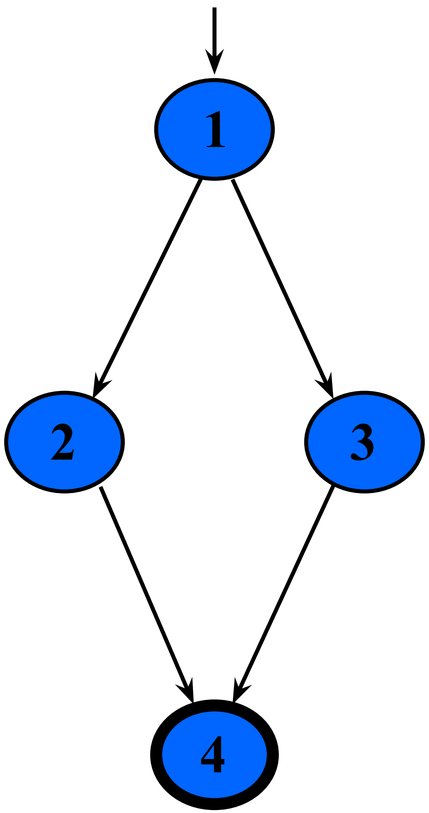


$N_0 = \{ 1 \}$

$N_f = \{ 1 \}$

$E = \{ \}$

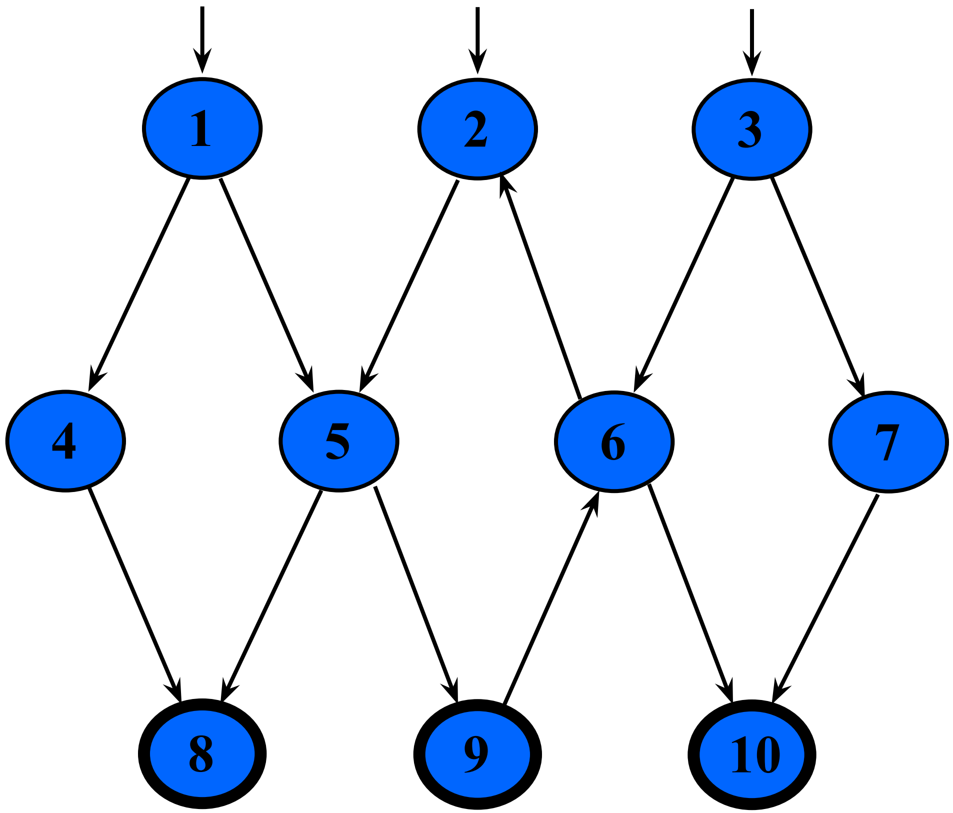
Example Graphs



N_0 = { 1 }

N_f = { 4 }

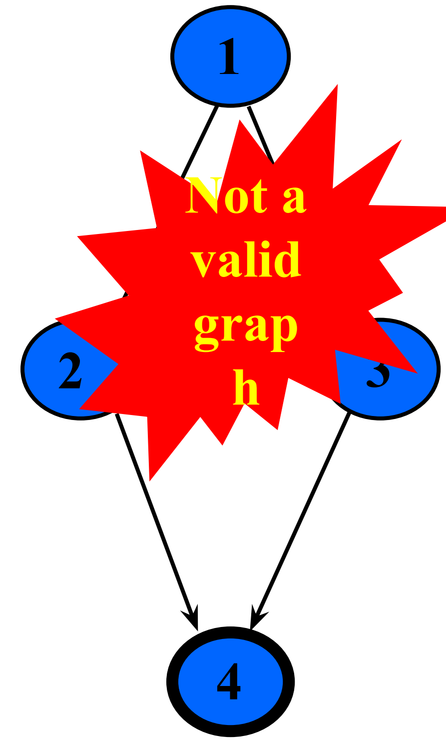
E = { (1,2), (1,3), (2,4), (3,4) }



N_0 = { 1, 2, 3 }

N_f = { 8, 9, 10 }

E = { (1,4), (1,5), (2,5), (3,6), (3, 7), (4, 8), (5,8), (5,9), (6,2), (6,10), (7,10) (9,6) }



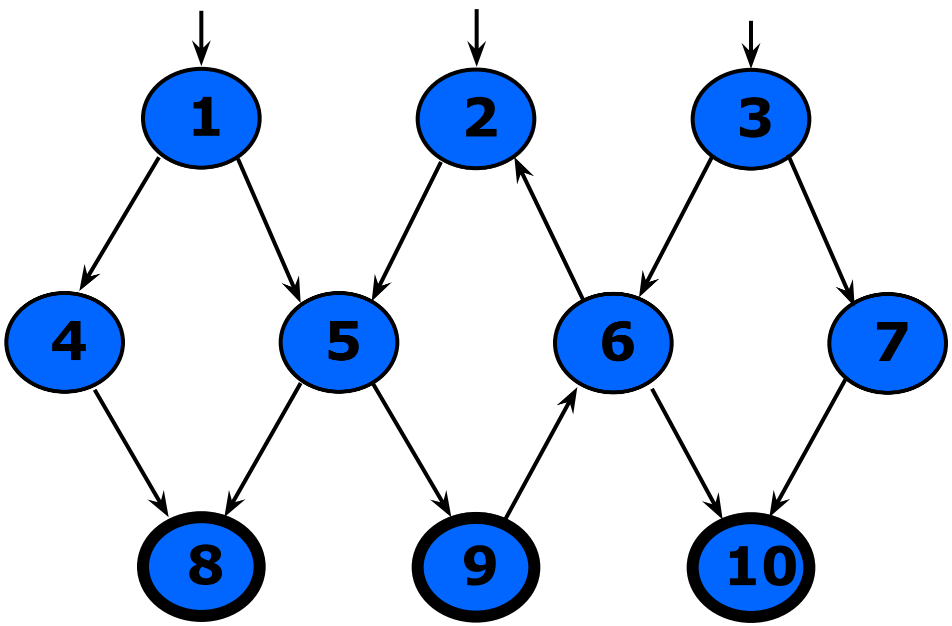
N_0 = { }

N_f = { 4 }

E = { (1,2), (1,3), (2,4), (3,4) }

Paths in Graphs

- Path : A sequence of nodes – [n_1, n_2, …, n_M]
  - Each pair of nodes is an edge
- Length : The number of edges
  - A single node is a path of length 0
- Subpath : A subsequence of nodes in *p* is a subpath of *p*
- Reach (*n*) : Subgraph that can be reached from *n*



A Few Paths

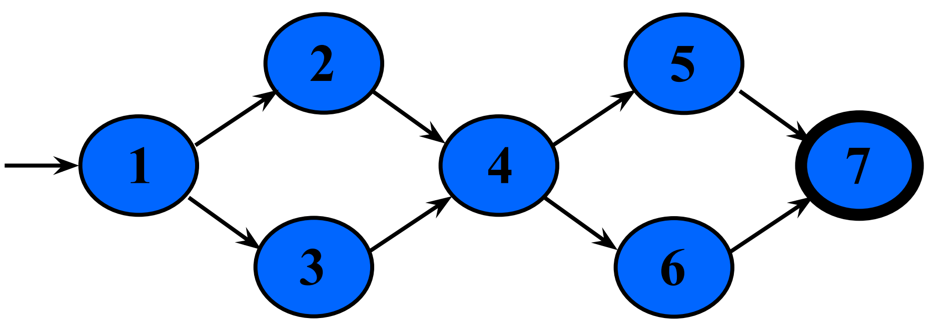
[1, 4, 8]

[2, 5, 9, 6, 2]

[3, 7, 10]

Test Paths and SESEs

- Test Path : A path that starts at an initial node and ends at a final node
- Test paths represent execution of test cases
  - Some test paths can be executed by many tests
  - Some test paths cannot be executed by any tests
- SESE graphs : All test paths start at a single node and end at another node
  - Single-entry, single-exit
  - N0 and Nf have exactly one node



<u>Double-diamond graph</u>

Four test paths

[1, 2, 4, 5, 7]

[1, 2, 4, 6, 7]

[1, 3, 4, 5, 7]

[1, 3, 4, 6, 7]

Visiting and Touring

- Visit : A test path *p visits* node *n* if *n* is in *p*
  A test path *p visits* edge *e* if *e* is in *p*
- Tour : A test path *p tours* subpath *q* if *q* is a subpath of *p*

Test path [1, 2, 4, 5, 7]

Visits nodes ? 1, 2, 4, 5, 7

Visits edges ? (1,2), (2,4), (4, 5), (5, 7)

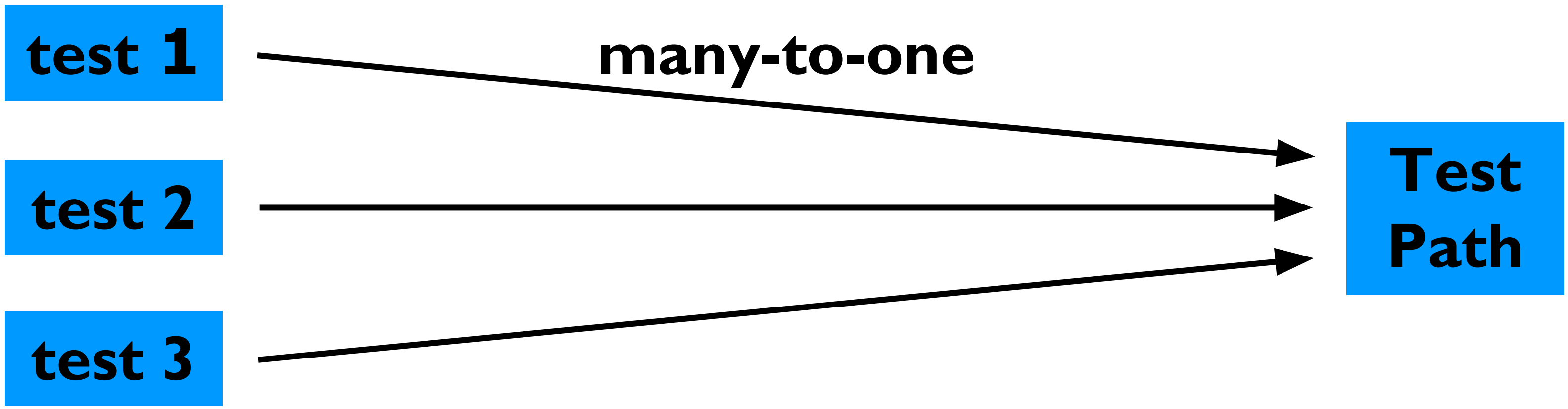
Tours subpaths ? [1,2,4], [2,4,5], [4,5,7], [1,2,4,5], [2,4,5,7], [1,2,4,5,7]

(*Also, each edge is technically a subpath*)

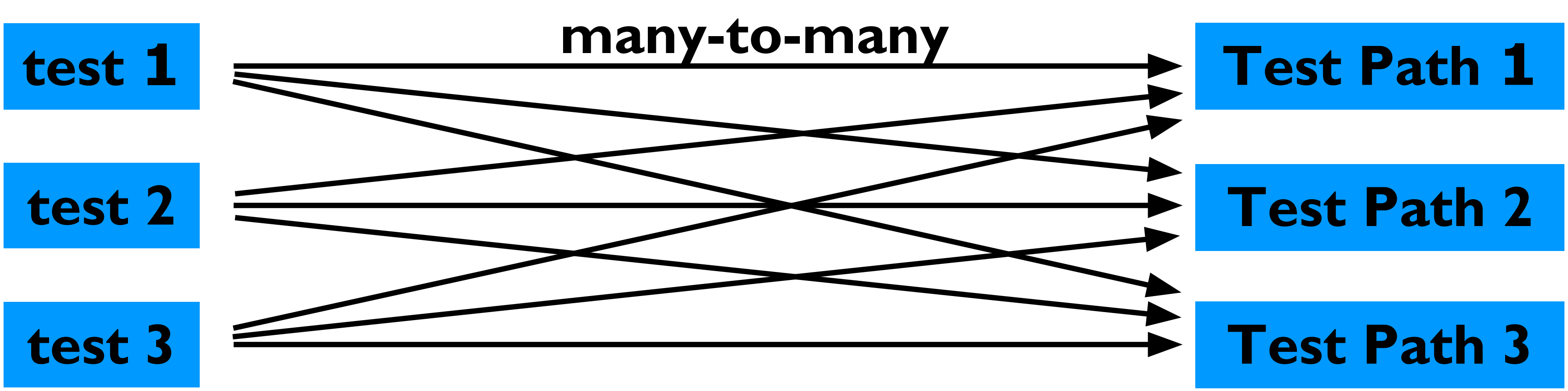
Tests and Test Paths

- path (*t*) : The test path executed by test *t*
- path (*T*) : The set of test paths executed by the set of tests *T*
- Each test executes one and only one test path
  - Complete execution from a start node to an final node
- A location in a graph (node or edge) can be reached from another location if there is a sequence of edges from the first location to the second
  - *Syntactic reach* : A subpath exists in the graph
  - *Semantic reach* : A test exists that can execute that subpath

Tests and Test Paths



Deterministic software–test always executes the same test path



Non-deterministic software–the same test can execute different test paths

Testing and Covering Graphs (7.2)

- We use graphs in testing as follows :
  - Develop a model of the software as a graph
  - Require tests to visit or tour specific sets of nodes, edges or subpaths
- Test Requirements (TR) : Describe properties of test paths
- Test Criterion : Rules that define test requirements
- Satisfaction : *Given a set TR of test requirements for a criterion C, a set of tests T satisfies C on a graph if and only if for every test requirement in TR, there is a test path in path(T) that meets the test requirement tr*
- Structural Coverage Criteria : Defined on a graph just in terms of nodes and edges
- Data Flow Coverage Criteria : Requires a graph to be annotated with references to variables

Node and Edge Coverage

- The first (and simplest) two criteria require that each node and edge in a graph be executed

<u>Node Coverage (NC)</u> : Test set *T* satisfies node coverage on graph *G* iff for every syntactically reachable node *n* in *N*, there is some path *p* in *path(T)* such that *p* visits *n*.

- This statement is a bit cumbersome, so we abbreviate it in terms of the set of test requirements

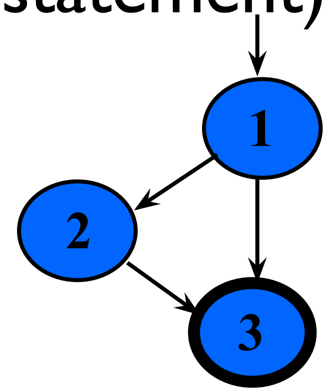
<u>Node Coverage (NC)</u> : TR contains each reachable node in G.

Node and Edge Coverage

- Edge coverage is slightly stronger than node coverage

> **Edge Coverage (EC) : TR contains each reachable path of length up to 1, inclusive, in G.**

- The phrase "*length up to 1*" allows for graphs with one node and no edges
- NC and EC are only different when there is an edge and another subpath between a pair of nodes (as in an "if-else" statement)

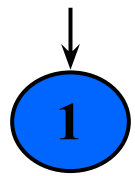


Node Coverage : ?TR = { 1, 2, 3 }
Test Path = [1, 2, 3]

Edge Coverage : ? TR = { (1, 2), (1, 3), (2, 3) }
Test Paths = [1, 2, 3]
[1, 3]

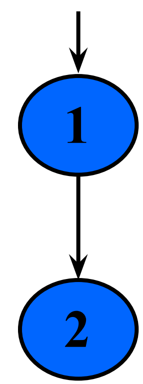
Paths of Length 1 and 0

- A graph with only one node will not have any edges



- It may seem trivial, but formally, Edge Coverage needs to require Node Coverage on this graph
- Otherwise, Edge Coverage will not subsume Node Coverage
  - So we define "length up to 1" instead of simply "length 1"

- We have the same issue with graphs that only have one edge – for Edge-Pair Coverage …

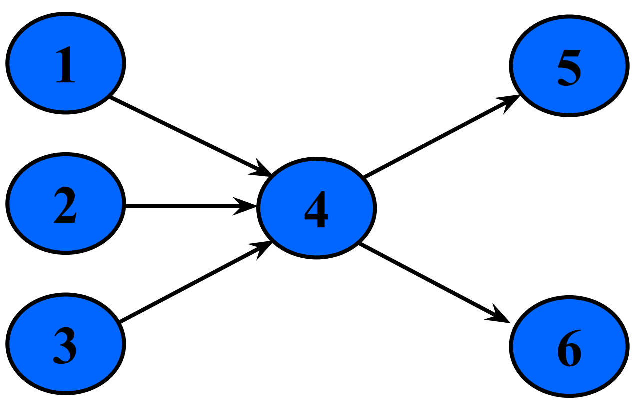


Covering Multiple Edges

- Edge-pair coverage requires pairs of edges, or subpaths of length 2

Edge-Pair Coverage (EPC) : TR contains each reachable path of length up to 2, inclusive, in G.

- The phrase "length up to 2" is used to include graphs that have less than 2 edges



Edge-Pair Coverage : ?

TR = { [1,4,5], [1,4,6], [2,4,5], [2,4,6], [3,4,5], [3,4,6] }

- The logical extension is to require all paths …

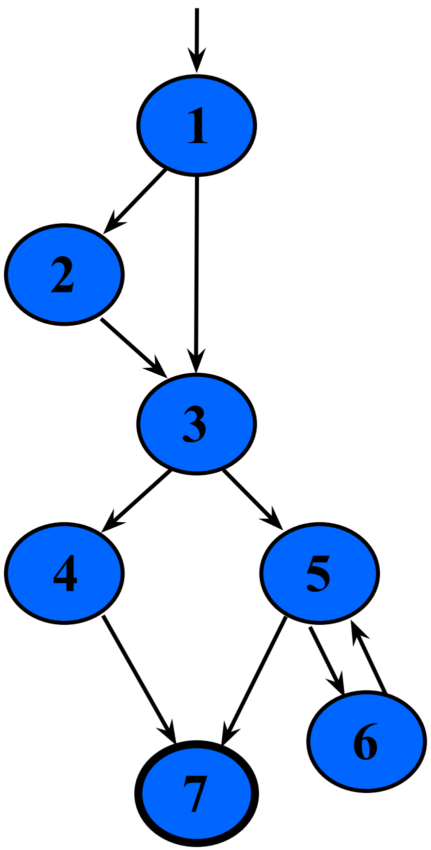
Covering Multiple Edges

Complete Path Coverage (CPC) : TR contains all paths in G.

Unfortunately, this is impossible if the graph has a loop, so a weak compromise makes the tester decide which paths:

Specified Path Coverage (SPC) : TR contains a set S of test paths, where S is supplied as a parameter.

Structural Coverage Example



Node Coverage

TR = { 1, 2, 3, 4, 5, 6, 7 }

Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 6, 5, 7]

Write down the TRs and Test Paths for these criteria

Edge Coverage

TR = { (1,2), (1, 3), (2, 3), (3, 4), (3, 5), (4,),
(6, 5) }

Test Paths: [1, 2, 3, 4, 7] [1, 3, 5, 6, 5, 7]

Edge-Pair Coverage

TR = { [1,2,3], [1,3,4], [1,3,5], [2,3,4], [2,3,5], [3,4,7],
[3,5,6], [3,5,7], [5,6,5], [6,5,6], [6,5,7] }

Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 7] [1, 3, 4, 7]
[1, 3, 5, 6, 5, 6, 5, 7]

Complete Path Coverage

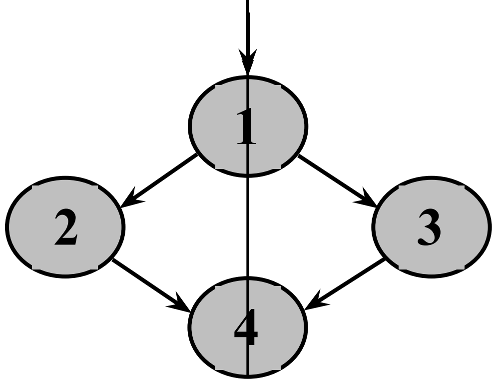
Test Paths: [1, 2, 3, 4, 7] [1, 2, 3, 5, 7] [1, 2, 3, 5, 6, 5, 7]
[1, 2, 3, 5, 6, 5, 6, 5, 7] [1, 2, 3, 5, 6, 5, 6, 5, 6, 5, 7] …

Handling Loops in Graphs

- If a graph contains a loop, it has an infinite number of paths
- Thus, CPC is not feasible
- SPC is not satisfactory because the results are subjective and vary with the tester
- Attempts to "deal with" loops:
  - 1970s : Execute cycles once ([4, 5, 4] in previous example, informal)
  - 1980s : Execute each loop, exactly once (formalized)
  - 1990s : Execute loops 0 times, once, more than once (informal description)
  - 2000s : Prime paths (touring, sidetrips, and detours)

Simple Paths and Prime Paths

- Simple Path : *A path from node n_i to n_j is simple if no node appears more than once, except possibly the first and last nodes are the same*
  - No internal loops
  - A loop is a simple path
- Prime Path : *A simple path that does not appear as a proper subpath of any other simple path*



Simple Paths : [1,2,4,1], [1,3,4,1], [2,4,1,2], [2,4,1,3], [3,4,1,2], [3,4,1,3], [4,1,2,4], [4,1,3,4], [1,2,4], [1,3,4], [2,4,1], [3,4,1], [4,1, ... 1,3], [2,4], [3,4], [4,1], [1], [2], [3], [4

Prime Paths : [2,4,1 ... ,4,1], [1,2,4,1], [3,4,1,2], [4,1,3,4], [4,1,2,4], [3,4,1,3]

Write down the simple and prime paths for this graph

Prime Path Coverage

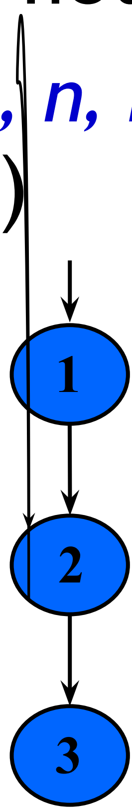
- A simple, elegant and finite criterion that requires loops to be executed as well as skipped

Prime Path Coverage (PPC) : TR contains each prime path in G.

- Will tour all paths of length 0, 1, …
- That is, it subsumes node and edge coverage
- PPC almost, but not quite, subsumes EPC …

PPC Does Not Subsume EPC

- If a node *n* has an edge to itself (*self edge*), EPC requires [*n*, *n*, *m*] and [*m*, *n*, *n*]
- [*n*, *n*, *m*] is not prime
- Neither [*n*, *n*, *m*] nor [*m*, *n*, *n*] are simple paths (not prime)



EPC Requirements : ?

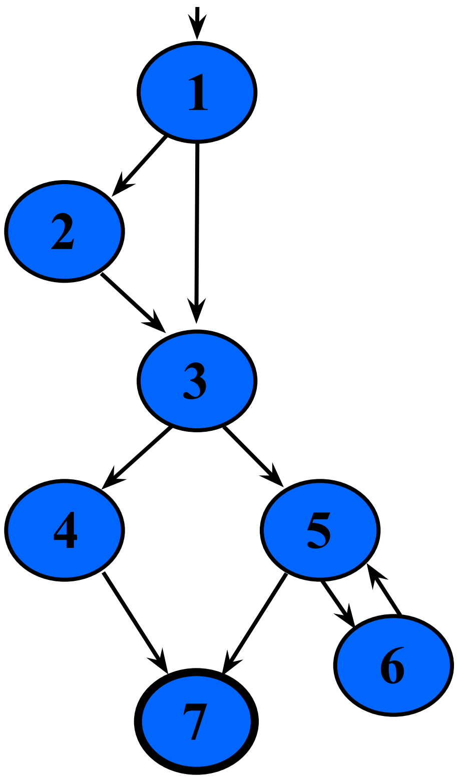
TR = { [1,2,3], [1,2,2], [2,2,3], [2,2,2] }

PPC Requirements : ?

TR = { [1,2,3], [2,2] }

Prime Path Example

- The previous example has 38 simple paths
- Only nine *prime paths*



Prime Paths

[1, 2, 3, 4, 7]

Write down all 9 prime paths

[1, 3, 4, 7]

[1, 3, 5, 7]

[1, 3, 5, 6]

[6, 5, 7]

[6, 5, 6]

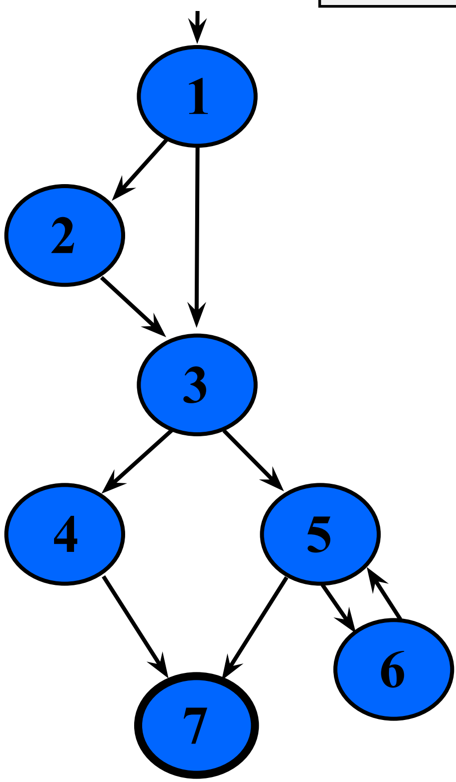
[5, 6, 5]

Execute loop 0 times

Execute loop once

Execute loop more than once

Simple & Prime Path Example



Simple paths

Len 0
[1]
Write paths of length 0
[5]
[6]
[7] !

'!' means path
es

Len 1
[1, 2]
[1, 3]
Write paths of length 1
[4, 7] !
[5, 7] !
[5, 6]
[6, 5]

Len 2
[1, 2, 3]
[1, 3, 4]
[1, 3, 5]
Write paths of length 2
[3, 5, 7] !
[3, 5, 6]
[5, 6, 5] *
[6, 5, 7] !
[6, 5, 6] *

Len 3
[1, 2, 3, 4]
[1, 2, 3, 5]
Write paths of length 3
!
!
[2, 3, 4, 7] !
[2, 3, 5, 6]
[2, 3, 5, 7] !

eans
ycles

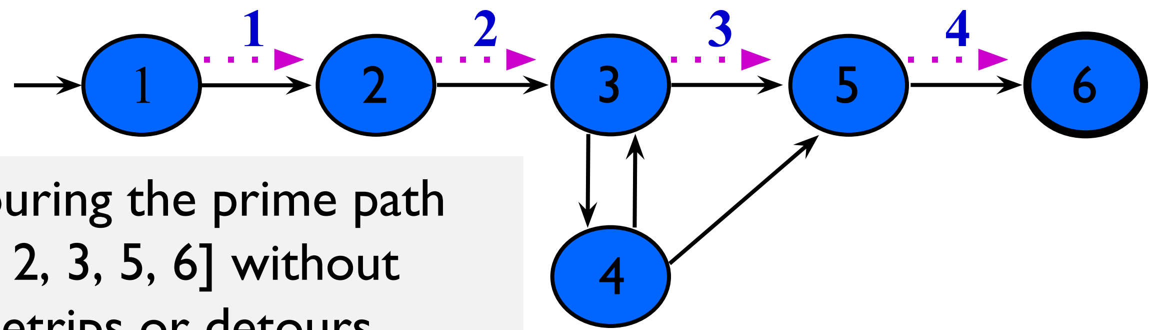
Len 4
[1 !
[1 !
[1
Write paths of length 4

Prime Paths ?

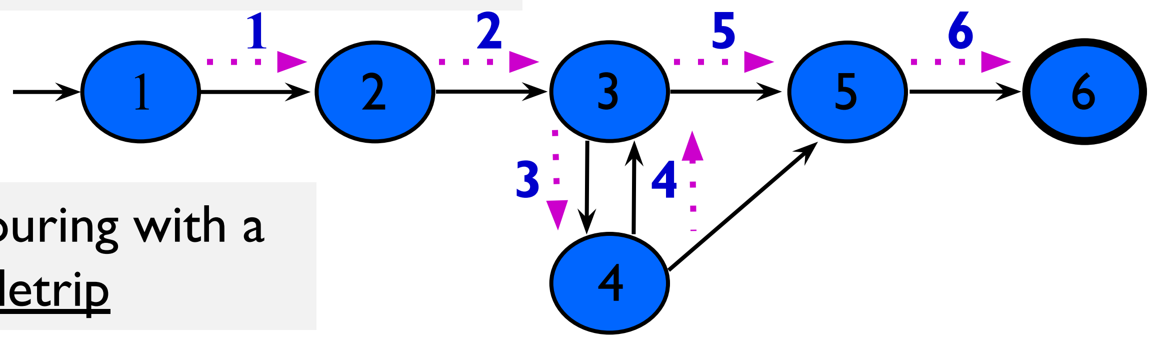
Touring, Sidetrips, and Detours

- Prime paths do not have internal loops … test paths might
- Tour : *A test path p tours subpath q if q is a subpath of p*
- Tour With Sidetrips : *A test path p tours subpath q with sidetrips iff every edge in q is also in p in the same order*
  - The tour can include a sidetrip, as long as it comes back to the same node
- Tour With Detours : *A test path p tours subpath q with detours iff every node in q is also in p in the same order*
  - The tour can include a detour from node *ni*, as long as it comes back to the prime path at a successor of *ni*

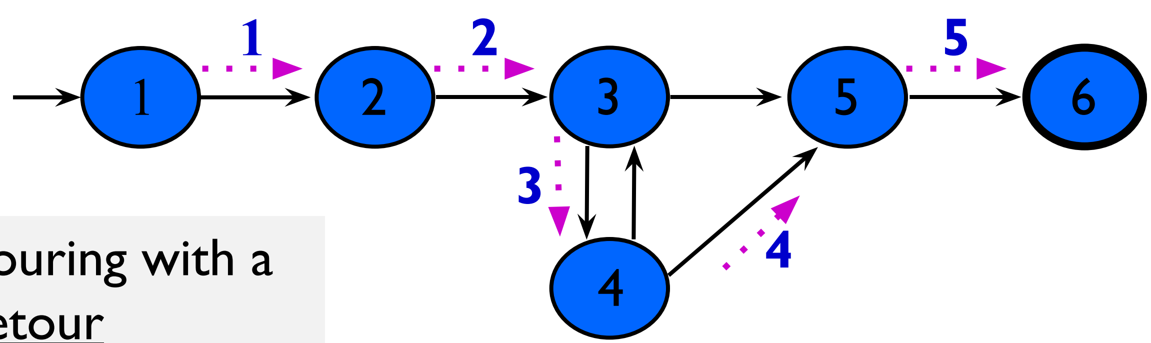
Sidetrips and Detours Example



Touring the prime path [1, 2, 3, 5, 6] without <u>sidetrips or detours</u>



Touring with a <u>sidetrip</u>



Touring with a <u>detour</u>

Infeasible Test Requirements

- An infeasible test requirement cannot be satisfied
  - Unreachable statement (dead code)
  - Subpath that can only be executed with a contradiction ($X > 0$ and $X < 0$)
- Most test criteria have some infeasible test requirements
- It is usually undecidable whether all test requirements are feasible
- When sidetrips are not allowed, many structural criteria have more infeasible test requirements
- However, always allowing sidetrips weakens the test criteria

Practical recommendation—Best Effort Touring

- **Satisfy as many test requirements as possible without sidetrips**
- **Allow sidetrips to try to satisfy remaining test requirements**

Round Trips

- Round-Trip Path : *A prime path that starts and ends at the same node*

<u>Simple Round Trip Coverage (SRTC)</u> : TR contains at least one round-trip path for each reachable node in G that begins and ends a round-trip path.

<u>Complete Round Trip Coverage (CRTC)</u> : TR contains all round-trip paths for each reachable node in G.

- These criteria omit nodes and edges that are not in round trips
- Thus, they do not subsume edge-pair, edge, or node coverage